Coursework 2 (Group) – Application of Computer Vision Techniques

Brief

This is a group coursework: please work in teams of four people.
Due date: Tuesday 6th May, 16:00
Handin: https://handin.ecs.soton.ac.uk/handin/2526/COMP6223/2
Required files: `report.pdf and code.zip`
Credit: 30% of overall module mark

Overview

Computer vision has been used in lots of fields and applications, with great performance. The goal of this assignment is to apply computer vision techniques on a practical problem – leveraging the techniques you have learned through the lectures (e.g. traditional techniques) and self-study (e.g. state-of-the-art techniques utilising deep learning).

Details

Data

Choose the problem and data to work on for example from:

- **Kaggle**: //www.kaggle.com/
- **Grand Challenges**: https://grand-challenge.org/challenges/
- **UK Biobank**: https://www.ukbiobank.ac.uk/

Computer vision technologies

You should try to develop/implement the best computer vision methods you can, and, in particular, you **MUST** include (and state it clearly in your report):

- Some of the traditional computer vision techniques introduced during the lectures;
- Some of the feature extraction techniques introduced during the lectures.

Code

You are free to choose the programming language you like.

The report

The report must be no longer than 5 pages (4 pages main content + 1 page reference) of A4 with the given Latex format, and must be submitted electronically as a PDF. The report must include:

- Your name and ECS user ID.
- A description of the data and problem, with data visualisation and reflection.
- A description of the implementation of the computer vision methods used.

- The validation of the computer vision methods on the data and problem.
- A discussion regarding the results obtained.
- We expect your report is written at a high standard academic level (i.e., the format and level at published top conferences in computer vision technologies).
- A short statement detailing the individual contributions of the team members to the coursework.

What to hand in

You are required to submit the following items to ECS Handin:

- Your 5-page report (as a PDF document in the CVPR format; max 5 A4 pages, no appendix).
- Your code enclosed in a zip file.

Marking and feedback

You will receive a grade out of 30 for this coursework. Marks will be awarded for:

- Successful completion of the task.
- Evidence of understanding.
- Well structured and commented code.
- Evidence of professionalism in the implementation and reporting.
- Quality and contents of the report.
- The quality/soundness/complexity of the computer vision approaches.

Standard ECS late submission penalties apply.

Individual feedback will be given covering the above points

Useful links

- Module COMP6223 website: https://secure.ecs.soton.ac.uk/module/2526/COMP6223/41230/
- Jupyter notebook: https://docs.jupyter.org/en/latest/
- https://www.datacamp.com/blog/computer-vision-projects
- https://blog.roboflow.com/computer-vision-projects/
- https://machinelearningprojects.net/computer-vision-projects/

Questions

If you have any problems/questions, use the Q&A channel on Teams.